Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MD Razia
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Roll Number	2520030385
Branch	CSE
Course Outcome	CO-2

Case Study Topic: HDFC NetBanking Daily Spend Prefix-Sum Management using Fenwick Tree (Binary Indexed Tree)

Work Done Summary:

In this case study, we implemented a Fenwick Tree (Binary Indexed Tree) to efficiently manage HDFC NetBanking customers' daily spending records.
The system stores daily transaction amounts and supports two key operations:

1. **Point Update** – Add a transaction amount to a specific day's spending record.
2. **Range Sum Query** – Calculate total spending within a given date range for monthly statement generation.

We constructed the BIT array from the daily spend data and used prefix sums to answer range-sum queries efficiently. The solution provides O(log n) time complexity for both updates and queries, making it suitable for large-scale banking systems.

Implementation Details:

1. Stored daily transaction amounts in the array **spend**[1..15].
2. Constructed the Fenwick Tree using cumulative range sums.
3. Each BIT node stores the sum of a specific range determined by LSB(i).
4. Calculated bit[4] by summing **spend**[1..4].
5. Calculated bit[8] by summing **spend**[1..8].
6. Calculated bit[12] by summing **spend**[9..12].
7. Used prefix sums to compute range queries: Range Sum(l,r) = Prefix(r) – Prefix(l–1)
8. Queried total spending from Jan 5 to Jan 12 using: Prefix(12) – Prefix(4)
9. Verified results with direct summation of daily transactions.
10. Observed logarithmic performance for updates and queries.

Result: Screenshots / Output

```
Run    Main
"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition\lib\idea_r:jar=56006
HDFC NETBANKING DAILY SPEND MANAGEMENT USING FENWICK TREE (BIT)
INPUT SPEND ARRAY (1..15):
Index:    1     2     3     4     5     6     7     8     9    10    11    12    13    14    15
Spend:  1200   800     0  2400  1500   600     0     0  3500     0  1100   950   700     0     0
FENWICK TREE (BIT) ARRAY (1..15):
Index:    1     2     3     4     5     6     7     8     9    10    11    12    13    14    15
BIT  :  1200  2000     0  4400  1500  2100     0  6500  3500  3500  1100  5550   700   700     0
bit[4]  = 4400
bit[8]  = 6500
bit[12] = 5550
VERIFICATION OF IMPORTANT BIT CELLS:
bit[4]  covers indices [1..4]   = 1200 + 800 + 0 + 2400 = 4400
bit[8]  covers indices [1..8]   = 1200 + 800 + 0 + 2400 + 1500 + 600 + 0 + 0 = 6500
bit[12] covers indices [9..12]  = 3500 + 0 + 1100 + 950 + 5550
PREFIX SUM CALCULATIONS:
Visited BIT cells for prefix(12): bit[12] bit[8]
prefix(12)  =  5550 + 6500  =  12050
Visited BIT cells for prefix(4): bit[4]
prefix(4)  = 4400
RANGE QUERY (JAN 5-12):
Spend[5..12] = prefix(12) - prefix(4)
             = 12050 - 4400
             = 7650
VERIFICATION (DIRECT SUM):
1500 + 600 + 0 + 0 + 3500 + 0 + 1100 + 950 = 7650  ✓
TOTAL SPEND FROM JAN 5 TO JAN 12 = ₹7650
PROCESS FINISHED WITH EXIT CODE 0
```

Fenwick Tree efficiently supported both update and query operations, making it highly suitable for banking transaction analytics and monthly statement generation.

GitHub Link: https://github.com/pondurutanvitha-stack/DSA-2

Student Signature	Faculty Signature